

MATHEMATICS
Introduction to Modern Algebra I
Review.

- (1) Define $A(S)$ and show that it is a group. If $|S| = n$, then show that $|A(S)| = n!$.
- (2) State the well ordering principle. State and prove Euclid's algorithm.
- (3) Define the greatest common divisor $c = (a, b)$ of two integers a, b (not both 0). Show that it exists and is unique. Prove that $c = ma + nb$ for some integers m, n . Compute $(104, 189)$.
- (4) Prove that there are infinitely many prime numbers.
- (5) Let p be a prime number. Prove that $\sqrt{p}$ is irrational.
- (6) If a, b are non-zero integers, show that ab equals the product of their least common multiple and their greatest common divisor.
- (7) Prove by induction that $\sum_{i=1}^n i^3 = \frac{1}{4}n^2(n+1)^2$.
- (8) Prove by induction that if $a \neq 1$, then $\sum_{i=1}^n a^i = (a^{n+1} - 1)/(a - 1)$.
- (9) Let z, w be complex numbers. Define $\bar{z}, |z|$, give a formula for z^{-1} , show that $\overline{zw} = \bar{z}\bar{w}$. When is $|z + w| = |z| + |w|$? State the triangle inequality.
- (10) find the solutions to $x^5 = 1 + i$.
- (11) write $\cos 4\theta$ in terms of $\sin \theta$ and $\cos \theta$.
- (12) Define a group; an abelian group; a cyclic group. Give examples of non commutative groups of order 10 and of infinite order; give an example of an infinite group all of whose elements have finite order.
- (13) Show that units and inverses are unique. Prove that $ab = ac$ implies $b = c$.
- (14) If all elements of G have order 2, show that G is abelian.
- (15) If $|G|$ is even, show that there is an element of order 2.
- (16) Define subgroups. Show that if H is a non-empty finite subset closed under multiplication, then it is a subgroup.
- (17) Define the center $Z(G)$ and the centralizer $C(g)$ and show that they are subgroups. Compute these in S_3 and D_4 .
- (18) If G is cyclic, show that it is abelian. What are the generators of G . Show that if a group has no proper subgroups, then it is cyclic.
- (19) Show that if G is abelian then $H = \{x \in G | x^2 = e\}$ is a subgroup, but this is not the case for all non-abelian groups.
- (20) Define equivalence relations. Show that if H is a subgroup of G then the relation $x \sim y$ iff $xy^{-1} \in H$ is an equivalence relation.

- (21) State and prove Lagrange's Theorem.
- (22) Define index of a subgroup.
- (23) If $|G| = m$ show that $x^m = e$ for all $x \in G$.
- (24) If $|G| = p$ and p is prime, show that G is cyclic.
- (25) Define Z_m and U_m . Prove that U_m is a group. Find the orders of elements in U_{18} , U_{20} .
- (26) Prove Euler's Theorem: If $(a, n) = 1$, then $a^{\varphi(n)} \cong 1$ modulo n .
- (27) Define left and right cosets. When is a subgroup normal? Show that subgroups of index 2 are normal.
- (28) Define homomorphisms, injective, surjective, isomorphism and Kernel. Show that if $\phi : G \rightarrow G'$ is a homomorphism, then $\phi(e) = e'$, $\phi(g^{-1}) = \phi(g)^{-1}$ and $\phi^{-1}(\phi(g)) = \text{Ker}\phi \cdot g$.
- (29) Show that ϕ is injective iff $\text{Ker}\phi = \{e\}$.
- (30) Show that $\phi(G)$ is a subgroup of G' which may not be normal. If ϕ is surjective and G is abelian, then G' is abelian. If G is cyclic then so is G' . Inverse images of (normal) subgroups are (normal) subgroups.
- (31) State and prove Cayley's Theorem.
- (32) Define normal subgroups and show that $\text{Ker}\phi$ is a normal subgroup.
- (33) Find all normal subgroups of D_4 and S_3 .
- (34) If N, M are normal and $N \cap M = \{e\}$, show that $nm = mn$ for all $n \in N$ and $m \in M$.
- (35) If N is normal in G define G/N and define a surjective homomorphism $\phi : G \rightarrow G/N$ (prove it).
- (36) State Cauchy's theorem and prove it for abelian groups.
- (37) Show that if N is normal in G then G/N is abelian iff $aba^{-1}b^{-1} \in N$ for all $a, b \in G$.
- (38) If G is abelian, $g, h \in G$ have orders n and m , then show that there is an element of G of order the least common multiple of n and m .
- (39) Find a surjective homomorphism from $(\mathbb{R}, \cdot)$ to $(\mathbb{Z}_2, +)$.
- (40) Show that if $G/Z(G)$ is cyclic, then G is abelian. (recall that $Z(G)$ is the center of a group.)
- (41) If m, n are relatively prime numbers, show that if an abelian group G , contains elements x and y of order n and m then it contains an element of order nm .
- (42) Do the previous exercise for non-abelian groups. (Hint: first show that $(x) \cap (y) = e$, then show that $xy = yx$ and conclude.)
- (43) State and prove the first homomorphism theorem.

- (44) Let $\phi : G \rightarrow G'$ be a surjective homomorphism. Show that the (normal) subgroups of G' are in 1-1 correspondance with the (normal) subgroups of G .
- (45) Show that there are no surjective homomorphisms $\mathbb{Z} \rightarrow \mathbb{Z}_4 \times \mathbb{Z}_6$, $\mathbb{Z}_8 \times \mathbb{Z}_3 \rightarrow \mathbb{Z}_4 \times \mathbb{Z}_6$, $\mathbb{Z}_4 \times \mathbb{Z}_6 \rightarrow \mathbb{Z}_8 \times \mathbb{Z}_3$.
- (46) Use Cauchy's theorem to show that a group with 99 elements has a normal subgroup.
- (47) Define the direct product of 2 groups. Prove that $\mathbb{Z}_4 \times \mathbb{Z}_3 \cong \mathbb{Z}_{12}$; show that $\mathbb{Z}_4 \times \mathbb{Z}_6 \cong \mathbb{Z}_{12} \times \mathbb{Z}_2$.
- (48) Let M, N be normal subgroups of a group G such that $M \cap N = \{e\}$ and $MN = G$. Show that $G \cong M \times N$.
- (49) State the theorem on the classification of finite abelian groups. Write all groups of order 10,000. How many distinct ones are there?
- (50) Find an element of order 70 in $\mathbb{Z}_{42} \times \mathbb{Z}_{45}$. What is the maximum order of an element in this group?
- (51) State the class equation and use it to show that if $|G| = p^n$ where p is a prime, then $Z(G) \neq \{e\}$.
- (52) Use the previous exercise to show that any group of order p^2 is abelian. Give an example of a group of order p^3 which is not abelian. What is $|Z(G)|$ for this group?
- (53) State Sylow's theorem.
- (54) Find all sylow subgroups in S_3 , D_4 , S_4 .
- (55) define $N(H)$ the normalizer of a subgroup H . Show that $N(H)$ is a subgroup of G and that H is a normal subgroup of $N(H)$.